Yuxin Lai

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Education

The Chinese University of Hong Kong Aug. 2026 – Present
Ph.D. Student in Computer Science and Engineering Advisor: Prof. Hong Xu Research Focus: AI Infra

Huazhong University of Science and Technology Sept. 2022 – June 2026
B.Eng. in Computer Science and Technology Score: 94.2 / 100 GPA: 4.83 / 5.00 Rank: **4 / 314 (Top 1.3%)**

Publications

KunServe**: Parameter-centric Memory Management for Efficient Memory Overloading Handling in LLM Serving**
Rongxin Cheng, **Yuxin Lai**, Xingda Wei, Rong Chen, Haibo Chen ***EuroSys'26*** (CCF-A) [Paper] [Code]

Internship & Research Experience

ByteDance Seed Infra Nov. 2025 – Present
Research Intern

Bitwise Precision Verification Framework

- Led development of a distributed training bitwise verification framework, designing systematic numerical alignment mechanisms for floating-point non-associativity and operator non-determinism across parallel setups.
- Built low-intrusion operator and collective communication interception in PyTorch, supporting bitwise alignment across TP, SP, PP, EP, FSDP, and diverse pre-training optimizations with minimal model code changes.
- Integrated precision-alignment skills with AI agents to deploy the framework into production training workflows, uncovering 10+ correctness bugs and cutting anomaly diagnosis from days of manual debugging to hours.

Pre-training Engineering Optimization

- Mitigated communication and memory-access bottlenecks in large-scale pre-training via compute/communication overlap scheduling and kernel fusion, hiding parallel communication overhead to improve throughput and MFU.
- Reduced peak GPU memory footprint in distributed training by combining selective activation checkpointing (SAC), activation offloading, and memory fragmentation mitigation to scale batch sizes and sequence lengths.

KunServe: LLM Serving System July 2024 – May 2025
Research Intern Advisors: Xingda Wei, Haibo Chen (**IPADS**, Shanghai Jiao Tong University)

- Developed the first parameter-centric LLM serving system that selectively drops replicated model weights under traffic spikes to instantly free GPU memory and eliminate queuing, leveraging CUDA virtual memory APIs for zero-kernel-intrusion physical memory reuse between parameters and KVCache.
- Implemented core system modules and multi-node GPU benchmarks; combined cross-instance pipeline execution, activation-prioritized network transfer, and lookahead micro-batch scheduling to eliminate bubbles, cutting P99 TTFT by up to **72.2**× and SLO violations by 7.2%–12.8% over vLLM, Llumnix, and InferCept.

DNN Checkpoint Sept. 2023 – Nov. 2023
Research Intern Advisor: Yu Hua (Huazhong University of Science and Technology)

- Built a multi-GPU fault-tolerant deep learning training platform, reproducing hierarchical asynchronous checkpointing across tiered storage (GPU–DRAM–NVMe) and hardware-software co-designed persistence to analyze checkpoint serialization and I/O bandwidth contention on training throughput.

Awards and Honors

CCF Elite Collegiate Award Oct. 2025
National Scholarship (2023–2024 Academic Year) Dec. 2024
National Scholarship (2022–2023 Academic Year) Dec. 2023
National Olympiad in Informatics in Provinces (NOIP), *First Prize* Dec. 2019
Lanqiao Cup National Software and IT Professionals Competition, *National Finals First Prize* June 2024
CCF Collegiate Computer Systems & Programming Contest (CCSP), *Bronze Award* Oct. 2023

Technical Skills

- **ML Systems**: Experienced in distributed LLM training and serving; proficient in PyTorch, CUDA, and NCCL.
- **Programming & Platforms**: Proficient in Python, C++, and Go; familiar with Linux, Docker, and Kubernetes.
- **AI Development Tools**: Skilled in AI Agent toolchains for efficient coding, debugging, and automated testing.
- **Languages**: IELTS 7.5 (C1 Advanced), with strong English reading, writing, and communication skills.